

DTSA 5011

Modern Regression Analysis in R

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Core	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 3 – Core	Machine Learning Engineer (64.3 / 100)	46.8 / 100
SPECIALIZATION Statistical Modeling for Data Science Applications		

47

CURRENT AVG SCORE
across 6 job roles

78

POTENTIAL SCORE
with recommended additions

+42

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Modern Regression Analysis in R provides foundational statistical modeling tools for data science. Introduction to methods, theory, and applications of linear statistical models, covering the topics of parameter estimation, residual diagnostics, goodness of fit, and various strategies for variable selection and model comparison. Attention will also be given to the misuse of statistical models and ethical implications of such misuse.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Machine Learning Engineer	64.3	100.0	+35.7	HIGH
Data Scientist	53.2	66.1	+12.9	LOW
ML Engineer	45.9	72.5	+26.6	MEDIUM
AI Engineer	41.8	71.1	+29.2	MEDIUM
Applied Scientist	40.8	79.2	+38.4	HIGH
MLOps Engineer	34.6	76.1	+41.6	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design 5 of 6 roles
systems

Deep Learning 1 of 6 roles
training

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

- 1** **HIGH** **Add System Design coverage** — required vocabulary in 5 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.
- 2** **LOW** **Optional: Deep Learning** — would broaden appeal to one additional role. Candidate terms: training.

GAP KEYWORDS BY ROLE

Machine Learning Engineer 64.3 → 100.0 (+35.7 pts)
predictive learning exploratory design development systems support complex

Data Scientist 53.2 → 66.1 (+12.9 pts)
social problems

ML Engineer 45.9 → 72.5 (+26.6 pts)
learning systems problems

AI Engineer 41.8 → 71.1 (+29.2 pts)
learning systems problems design make

Applied Scientist 40.8 → 79.2 (+38.4 pts)
biology computational problems learning systems design real development

MLOps Engineer 34.6 → 76.1 (+41.6 pts)
systems learning training development error

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.